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Radio Frequency Identification (RFID) Technology Facilitates Enhanced Acid Stimulation Coverage for Increased Oil Recovery Factor in Deepwater Multi-Zone Intelligent Well

C. Munro, Weatherford, Aberdeen, Scotland; A. Gomes, Weatherford, Macae, Brazil; D. Nunes e Silva, F. Salatiel de Oliveira, and A. Barreto Coutinho, Neto, Petrobras, Rio de Janeiro, Brazil

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Abstract

Targeted acid stimulation of high productivity reservoirs through intelligent well systems have historically been constrained due to limited tubing hanger penetrations and by the intelligent well systems themselves, considering available configuration flow area and zonal flow control. Besides the carbonate reservoirs being heterogeneous by nature, the long extension of the producing intervals makes the task of distributing acid treatment even more challenging. This paper will focus on the integration of remotely operated sliding sleeves into deep-water three-zone intelligent wells to enhance stimulation coverage without compromising future production control.

Geological analysis was conducted to assess the benefits of increased treatment coverage during acid stimulation. The RFID stimulation sleeves were placed in the production zone of the completion alongside the ICV (Interval Control Valve). The sleeves were run closed and once the completion was set and tested and ready for acid stimulation A pressure pulse signature was applied to activate the electronic trigger and open the sleeve. The stimulation was applied through both the RFID sleeve and ICV. After treatment the RFID sleeve was closed by pumping RFID tags into the well and a successful pressure test confirmed valve closure and well integrity.

The inclusion of a multi-cycle remotely operated electronic sliding sleeve into the intelligent well design increased the available acid injection area, in combination with the ICV's and without using any additional tubing hanger penetrations, to enhance overall production across the well campaign to increase the Oil Recovery Factor (ORF). Additionally, the operational flexibility; using RFID tags, remote pressure cycles, and timers allowed the operator to selectively target and treat zones during acid stimulation, whilst having on-demand contingency options to significantly reduce the likelihood and risk of intervention. Whilst this campaign utilized one RFID sliding sleeve per well, there are no limitations to the number of sleeves that can be placed in any well.

This technology milestone demonstrates the possibility to significantly expand the treatment coverage available to the reservoir during acid stimulation. For example, multiple RFID sleeves could be placed into each zone that contains one ICV; the RFID sleeves would be used for acid stimulation and the ICV for

production control, which ultimately delivers operational flexibility, increased oil recovery and therefore value. Furthermore, after the stimulation has been performed, the RFID valve could act as a mechanical contingency for the ICV during well production.

Electronic Sliding Sleeve with Integrated RFID Technology

Sliding sleeves have been an integral and critical part of well completion design for many years. These have typically taken the form of mechanically operated by shifting tools deployed on slickline, wireline or washpipe to allow production or injection from or to targeted zones of the reservoir. In high-cost operating environments, particularly deepwater offshore, intervention with slickline, wireline or coiled tubing is expensive and presents elevated risks that are not always acceptable.

RFID technology within downhole completions products has been available on the market for many years. However as the technology has matured, the range of devices that can be controlled using RFID has expanded allowing new opportunities and applications to open up to integrate the RFID technology to enhance existing methods and practices in oil recovery.

The RFID electronics sliding sleeve (eSSD) presented in this paper was specifically designed after successful award of the campaign by the client. Designed to fit inside 9-5/8" casing with 4-1/2" end connections, it has a concentric design with integrated by-pass slots to hydraulic lines to operate the ICV's, chemical injection lines and electrical gauges cable within the well.

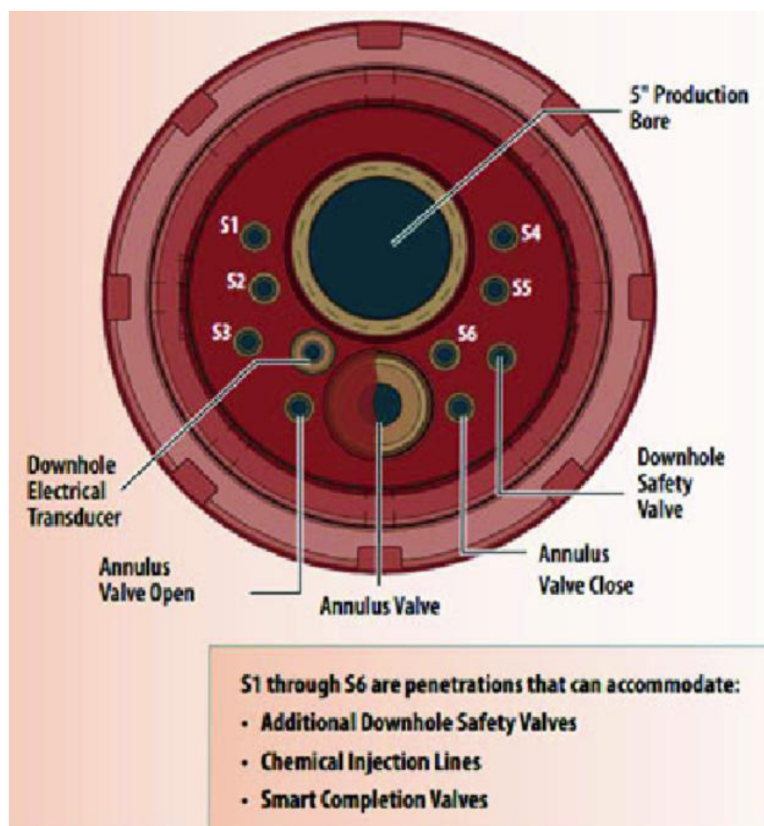


Figure 1—Example of Typical Tubing Hanger Schematic

How the sleeve works

The eSSD comprises two main sub-systems; the RFID controller featuring battery pack, micro hydraulic pump, fluid control valve, hydraulic fluid system and RFID antenna. And the mechanical sliding sleeve; featuring inner sleeve, flow port housing and non-elastomeric multi-cycle sealing system.

The sleeve can be run in the well in either the open or closed position, closed being most common. To open the sleeve a pressure pulse sequence is applied to the well from the surface, usually against some form of plugging device below. Once the command has been received the tool carries out the desired operation by moving the control valve and then starting the on-board micro pump. The pump drives the sleeve open, if its initial state was closed, until it reaches the end of the stroke and the pump detects the termination pressure. When the sleeve is open, injection and or production can take place through the flow ports. After any period of flow, the sleeve can then be re-closed by pumping RFID tags to the sleeve. As it passes through the tool's antenna, the instruction that has been pre-programmed on the RFID tag is decoded to the on-board electronics and the closing of the sleeve begins.

The design of this system offers many benefits over conventional sliding sleeves in that no control lines to surface are required, the stroking force available from the on-board micro pump can deliver up to 40,000lbs every time and additional contingency options to actuate the sleeve are always available, such as timers and secondary pressure cycles.

Operational Key Features

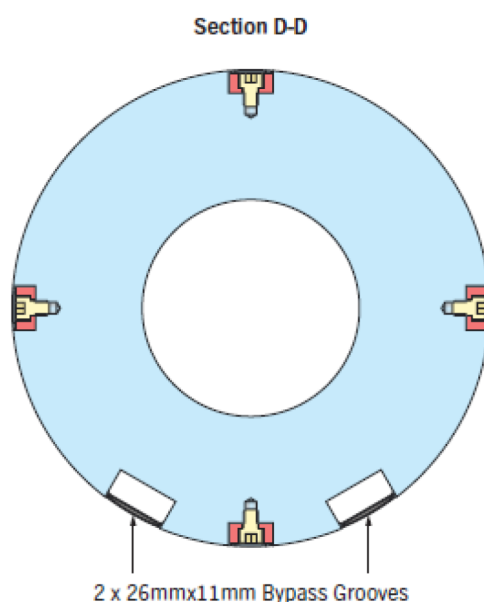


Figure 2—eSSD Control Line By-pass Section View

Specification of eSSD

Table 1—Specification of eSSD

Parameter	Value
Size	9-5/8" x 4-1/2"
Pressure Rating	7500 Psi
Battery Life	1 year
Special Features	2 x 26mm x 11mm by-pass slots
Temperature Rating	4-150 deg C

Trigger Methods



Figure 3—RFID Tags & Carrier

The RFID tags used in the eSSD and other completions tools are cylindrical versions of the same chip used in multiple contactless credit cards, Toll passes and other such systems. They are placed in a silicone jacket with a flow-head to keep them intact and working while traveling downhole. These tags can be dissolvable if there is any clogging concern.

Electro-Hydraulic Sub-system

The electro-hydraulic system is completely isolated from any well fluids providing a clean operating environment for the pump, fluid control valve and pressure transducers. Additionally, the fluid system is filtered to minimum NAS 6 and operates as ‘net-zero’; no fluid is lost or gained during operation.

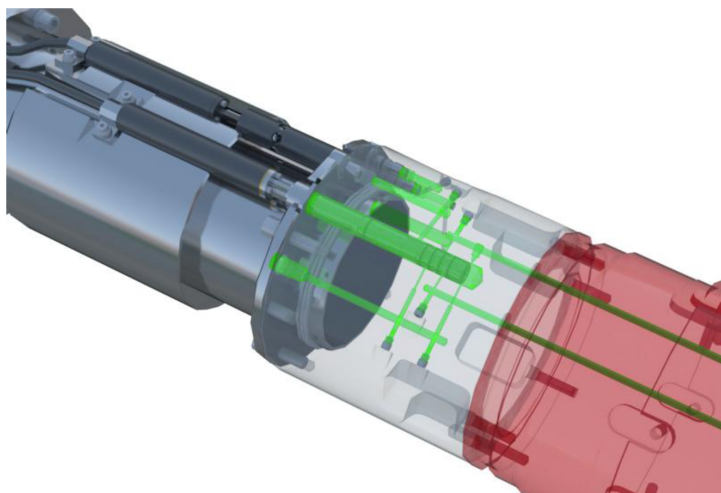


Figure 4—Hydraulic Manifold

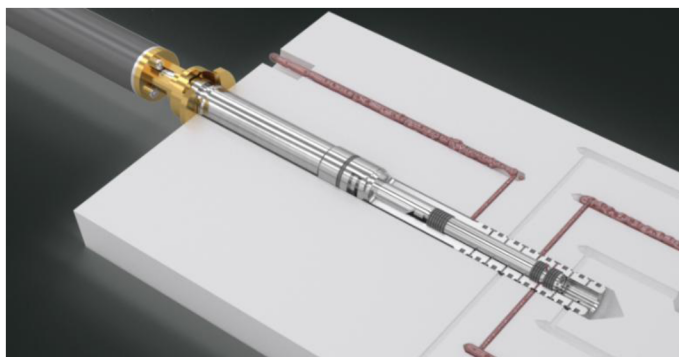


Figure 5—Fluid Control Valve & Micro Pump

Sleeve Sub-section Key Features

The sleeve section is constructed from three simple housings with an inner sliding sleeve. This sleeve features diffused opening to provide longevity to the non-elastomeric sealing system. That sealing system is engineered to withstand exposure to a range of standard and customized oilfield chemicals, liquids and gases. The flow area through the sleeve is larger than the tubing restriction thereby reducing any concern about significant erosion.

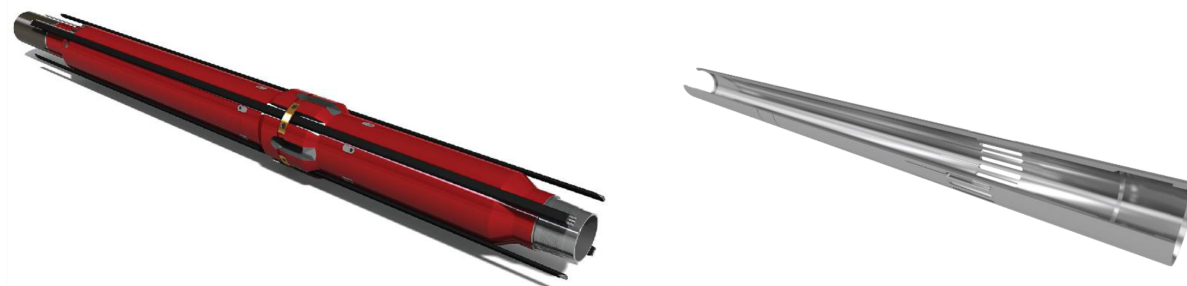


Figure 6—eSSD Sleeve Section

Qualification Testing

The eSSD was qualified to API 19AC + client specific requirements to ensure product integrity, reliability and longevity. A summary of the test programme is given in [Table 2 - eSSD Testing Summary](#) below.

Table 2—eSSD Testing Summary

Test Description	Objectives	Outcome
Long Term Battery Testing	2000 hours + minimum 6 sleeve functions	Successful function after 2000 hours and 6 sleeve actuations
Operating Temperature Battery Test	Maximum number of sleeve functions	40 functions
High Temp Battery Test 150 C	Minimum 3 x cycles of sleeve	83 x sleeve functions at max temp
Electronics & Hydraulic	Minimum 10 Thermal Cycles	10 x thermal swings + 12,000psi function
Electronics & Hydraulic Vibration & Shock	10g RMS random vibration, 10-600 Hz with spectrum: +3dB/octave from 10-100 Hz, constant from 100-500 Hz and -3dB/octave from 500-600 Hz Shock Test - Apply three impact pulses (Shock Test) to the equipment, with 30 g acceleration, 11 ms half sine, in all directions of each of the three perpendicular axes (X, Y, Z).	Successful function tests after all vibration and shock testing.
API 19AC Differential opening test	3 x open close cycles at 1500psi tubing to annulus at 125 deg C	6 successful unloads
API 19AC Envelope Testing	Test Points around envelope	All points tested + additional thermal swing + 3 x open close cycles
Assembly, FAT and Quality Plan Checks	Confirmation of production assembly & FAT	Successful assembly with minor procedure improvements.
Ambient electronic function test	Full operating logic with RFID tags and pressure pulses	Successful tests
Maximum Hydrostatic Testing	Sleeve function at 16,500psi	Successful tests + final function
Contingency Shifting Tool Test	Confirm override option pre & post hydraulic disconnect	Pass

Strategic Placement of eSSD for Optimized Stimulation

Producers and injectors wells in Pre-Salt offshore fields commonly benefit from up to three production intervals isolated with feed through packers and flow is controlled via Interval Control Valves (ICV's) as per reservoir engineering protocol. The hydraulic operating lines for the ICV's, chemical injection and electrical

cable consume all the remaining tubing hanger feed-through slots, thus the scope to operate additional sliding sleeves via control lines is not available. Remotely operated sleeves are the only option to allow acid treatment and injection from another portion of the interval without through tubing intervention.

Standard PACI 3Z architecture is to place the Interval Control Valve (ICV) at the top of each zone as shown in Figure 7.

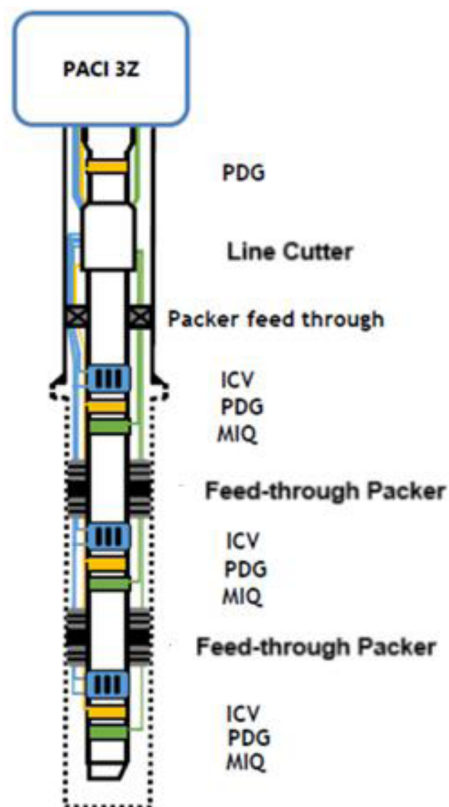


Figure 7—Example PACI 3 Zone Schematic

After well drilling and reservoir logging, the subsea and completion experts from the operating company assess porosity, permeability and image logs. The number and extension of zones is defined depending on geological features and reservoir management objectives. Stimulation experts then evaluate available data to define whether having an extra valve for acid injection in any of the intervals is worthy or not. An in-house matrix acidizing software can be used to compare simulated post acid skin considering single or dual injection points. Permeability distribution along the interval and zone extension are among the key parameters that impact on results and lead to defining eSSD depth and stimulation strategy to achieve optimized zonal coverage, lower skin and higher productivity/injectivity.

The eSSD is typically positioned at the bottom of the interval or slightly above it, depending on well logs and interval extension. For instance, if there are high permeability layers at the base of the interval, it may be more advantageous to position the second valve above these layers rather than strictly at the bottom of the zone. Additionally, there are various acid pumping strategies that can be employed, such as:

1. Pumping the first part of the planned acid volume through the eSSD and then the remaining volume through the ICV, located at the top of the interval; or
2. Pumping part of the acid volume from the top of the interval (ICV) and finishing the injection from the bottom of the interval (eSSD).

Alternatively, the entire designed acid sequence can be pumped simultaneously through both ICV and eSSD, depending on the scenario.

In the presented field case, the eSSD was strategically positioned at the base of a long interval (approximately 200 meters), where the formation exhibits lower permeabilities. A minor portion of the acid treatment—about 18% of the reactive volume—was initially pumped from the top of the interval. Afterward, the ICV was closed, and the main phase of the treatment was injected via the eSSD, targeting the lower section of the interval. The acid stimulation stages involved an alternating sequence of acid solutions, viscous diverging agents, and emulsified acid. This strategy was designed to avoid preferential treatment of the upper sections, considering the significant permeability contrast and the length of the interval. An in-house stimulation software played a crucial role in defining the optimal approach, including the selection of acid systems, stage volumes, and the valve operation sequence.

After matrix acidizing, an injectivity test was performed to evaluate post treatment well performance. Completion fluid was pumped at 20 bpm rate, followed by a falloff period. The analyzed data confirmed the success of the operation: the interval achieved a skin factor of -4.6 and an impressive injectivity index of 767 (m³/d)/(kgf/cm²).

Considering that PACI 3Z configuration converges the flow to a single valve, running a Production Logging Tool (PLT) to evaluate flow distribution behind the production string would be ineffective. Instead, a temperature log was conducted as a qualitative tool, allowing the team to confirm that the entire interval was successfully exposed to acid.

Figure 8 presents two logs side by side: permeability (on the left) and post-acid temperature (right). The high-permeability layers at the top of the interval are highlighted in yellow. Although the reaction of hydrochloric acid (HCl) with calcium carbonate (CaCO₃) is exothermic, injected fluid temperature is significantly lower than bottomhole static temperature due to the high water depth. The dashed red line represents an approximation of the thermal gradient. Deviations to the left of this line indicate a cooldown, resulting from the higher volume and/or local rate of the injected fluid. Therefore, the local temperature distortions to the left may be directly associated with the influx of acid, clearly demonstrating that the entire green-highlighted area was exposed to acid flow.

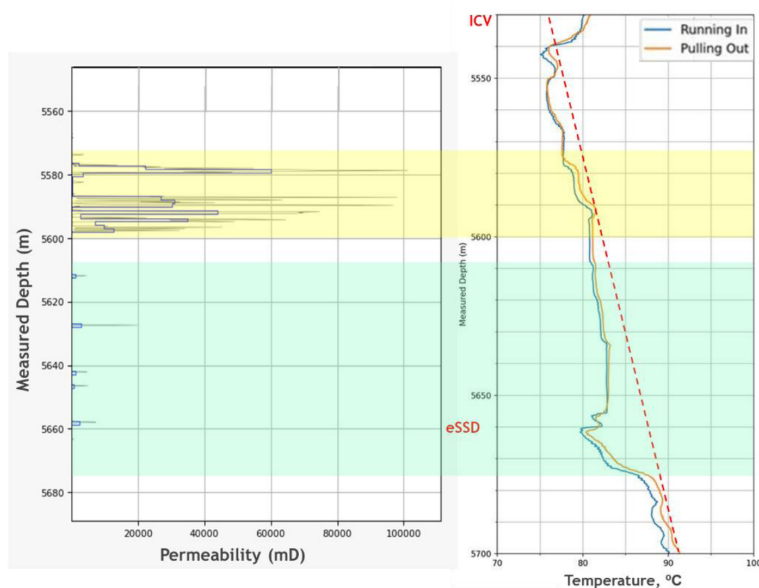


Figure 8—Permeability log (left) and post acid temperature log (right). Temperature deviation from the thermal gradient (dashed red line) indicates there was significant fluid flow along the whole interval.

If the entire acid job had been pumped solely through the ICV, a portion of the yellow-highlighted interval would likely have taken the acid preferentially, even with high injection rates and the aid of chemical diverters. By initiating injection from the bottom of the interval (eSSD), it was possible to ensure better acid distribution all through the green-highlighted extent. Although the graph scale suggests low permeabilities at the green-highlighted portion of the reservoir, several peaks above 1 Darcy that were certainly damaged during drilling and would have impaired production had acid injection been performed conventionally. Therefore, although temperature log does not provide a quantitative distribution of flow rates, the interpretation of the graph allows to conclusively determine that there was flow along the entire interval and that the stimulation strategy was successful, in terms of zonal coverage and therefore excellent well productivity.

A study considering a Pre-Salt field scenario estimated that the impact of different acid distribution and effectiveness on the Recovery Factor of the fields can be significant, leading to an increase of around 2.5%, based on a baseline case of permeability anisotropy (Silva et al., 2020). On the other hand, considering high anisotropy between horizontal and vertical permeabilities, the difference in the Recovery Factor can reach up to 7 percentage points. Considering field lifetime, the potential benefit justifies investments in different strategies to enhance stimulation results.

CFD for flow conditions and erosion prediction

Computation Flow Dynamic (CFD) studies were conducted to understand the impact of injection and production flow on the eSSD.

Given that these sleeves can be used for short-term stimulation or long-term production of hydrocarbons for decades, understanding the limitations and using the most up-to-date engineering analysis methods was critical to predicting life of the product down hole. These simulations must assess whether the RFID valve could serve not just for stimulation scenarios as but also as a replacement for the ICV in the event of failure during production or injection in ultra-high-flow wells, particularly in producing wells with fine solids.

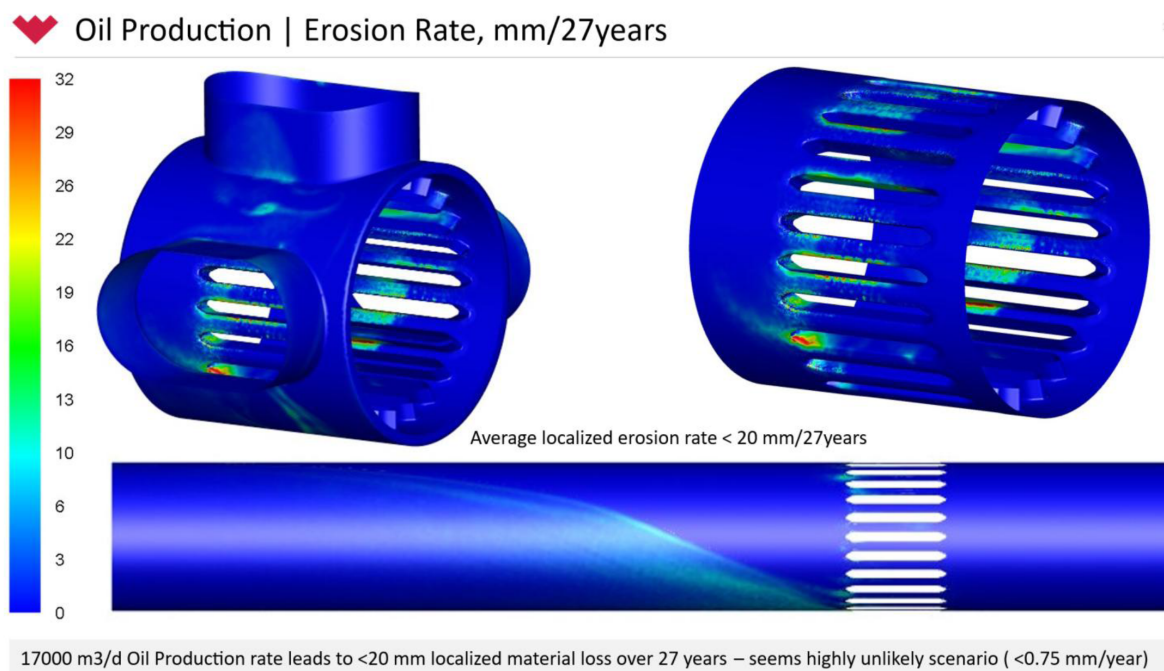


Figure 9—CFD Erosional Model Analysis

As can be seen from Table 3 - CFD Inputs and Results, considering a production rate of 170,000 m3/day expected life of the sleeve can be stated as 27 years, with a total reduction of material less than 20mm.

Table 3—CFD Inputs and Results

Scenario	Fluid	Fluid	Solids	Max V, ft/s	ΔP , psi	Avg Erosion Rate
Injection	Water 1025 kg/m ³ 0.97cP	12400 m ³ /d	n/a	130	70	n/a
Injection	CO ₂ Properties = f(P, T)	4.5 MM Sm ³ /d	n/a	114	60	n/a
Production	Oil 7.5ppg 0.41cP	17000 m ³ /d	2650 kg/m ³ 0.001 wt.% d = 50 micron	180	175	< 20 mm/27years

17000 m³/d Oil Production rate leads to <20 mm localized material loss over 27 years – seems highly unlikely scenario (<0.75 mm/year)

RFID Logic Pre-job Planning

Of such benefit with RFID and electronic based completion tools is the ability to design application-specific operating logic to overcome any restrictions or simply enhance the efficiency of the operation. With this benefit comes additional layers of options available to the user. With this in mind, pre-job planning is of the utmost importance and requires detailed analysis prior to running the tools in a well. To ensure the eSSD was configured in the most beneficial and efficient way, a complete logic flow chart was constructed detailing the operational steps along with the available contingency options at each stage. This is shown in Figure 10 - eSSD Application Logic Flow Chart

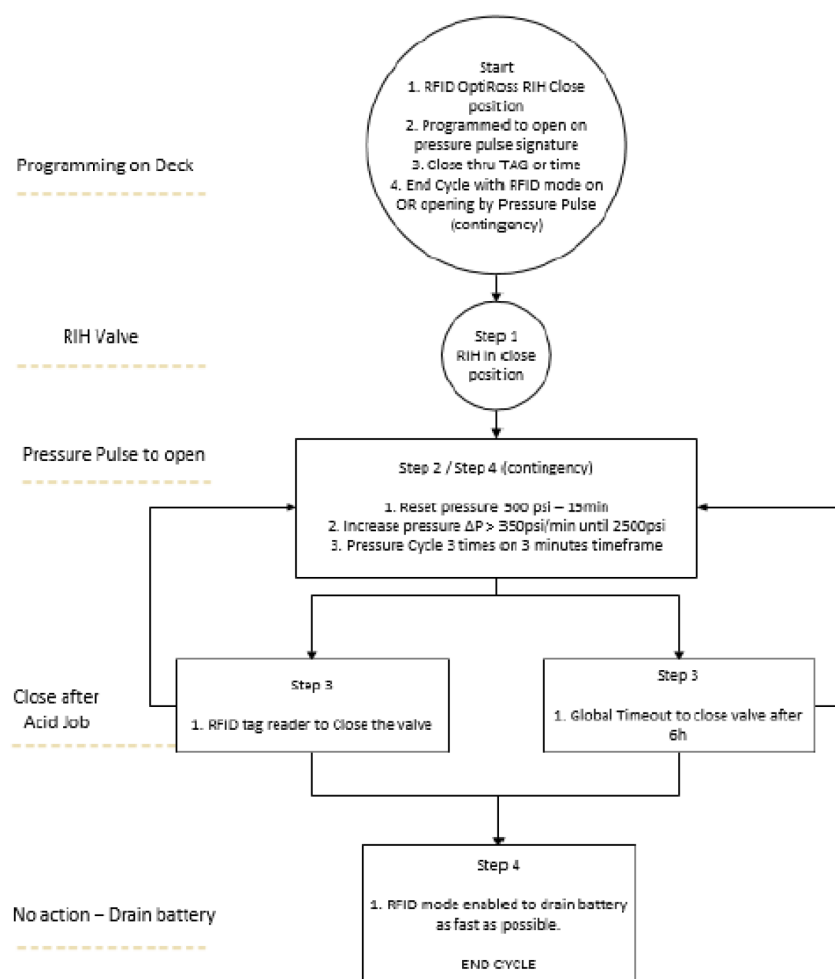


Figure 10—eSSD Application Logic Flow Chart

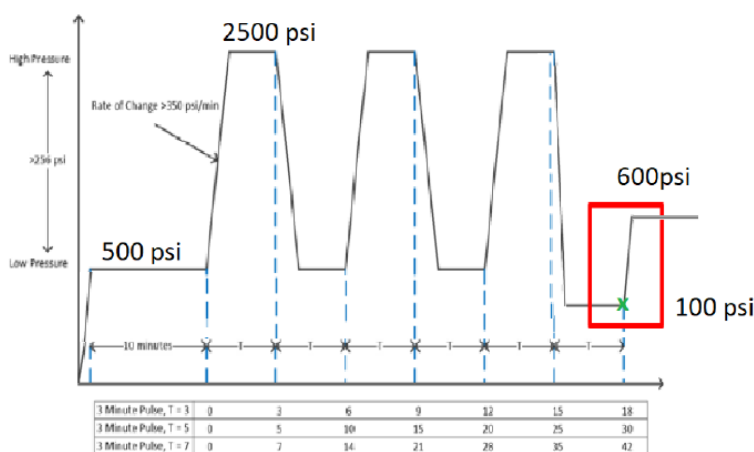


Figure 11—Pressure Pulse Sequence Chart

After successful land off at target depth, the pressure pulse is applied to the well which commands the sleeve to open. The ‘opening pressure signature’ can be left at any value, depending on the application. This is highlighted in the red box above. This opening function, when complete, initiates a mode change within the tool from pressure pulse detection mode into RFID tag detection mode, which is done by powering up the antenna. This also activates the background timer to start counting, which was set at 6 hours.

Once the acid injection operation is nearing completion, the RFID tags are introduced to the flow to command the sleeve to close. Finally, the tool was set up to drain the full battery to ensure its final status.

Experiences with valves employing the same remote cycling control system indicate that proper logic programming, combined with pre- and post-download verification procedures, should anticipate and avoid potential human errors.

eSSD Operational Deployment

After drilling and reservoir evaluation was complete and the decision had been made to deploy the eSSD on this well and in the particular location of the reservoir, the completion was successfully run to TD. A three-minute pressure sequence was applied against the isolation bull plug device at bottom which commanded the eSSD to open. A pressure drop at surface indicated a successful operation, as shown in Figure 13 - eSSD Opening Sequence from Well. This pressure pulse also activated a back-up timer to close the eSSD after 6 hours in case issues occurred and circulation out the sleeve was not possible.



Figure 12—eSSD Sleeve in Closed Position at RIH

eSSD Opening Sequence

Upon completion of the acid stimulation operation, RFID tags were launched from a surface assembly through a dedicated launching valve. These tags were programmed to close the eSSD immediately. A final pressure test was applied against the closed sleeve confirming valve closure and well integrity in preparation for production.

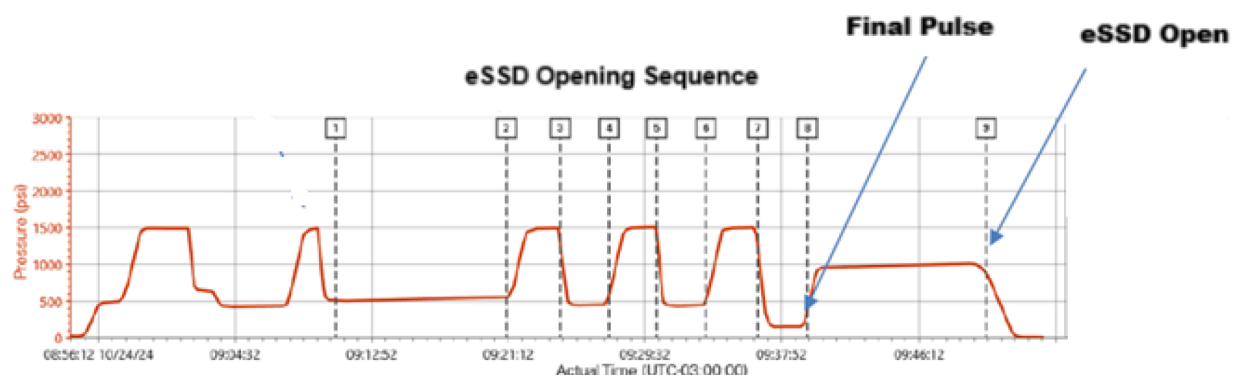


Figure 13—eSSD Opening Sequence from Well

Results

The installation of the eSSD, strategically positioned in high-permeability carbonate wells, has enabled the operator to achieve enhanced stimulation results and productivity. Although running a PLT for quantitative evaluation of flow distribution is not possible in the three-zone intelligent completion configuration, the success of the strategy has been verified by conducting temperature logs and performing post-acid formation evaluations (injectivity test and fall off analysis).

Seven wells have been completed using the eSSD as an additional acid injection point and no operational failures have been observed so far. Depending on permeability distribution, the operator considers the possibility of running up to 1 valve per zone with no distinction between producer or injector well.

Conclusions

The inclusion of a multi-cycle remotely operated electronic sliding sleeve into the intelligent well design in combination with the ICV's and without using any additional tubing hanger penetrations has enhanced stimulation results and increased long-term production. Additionally, operational flexibility; using RFID tags, remote pressure cycles, and timers has allowed the operator to selectively target and treat zones with acid stimulation, whilst having on-demand contingency options to significantly reduce the likelihood and risk of intervention. Whilst this campaign utilized one RFID sliding sleeve per well, there are no limitations to the number of sleeves that can be placed in any well.

This technology milestone demonstrates the possibility to significantly expand acid treatment zonal coverage during acid stimulation, which is always a challenging goal in high permeability, heterogeneous and long extension intervals, common in the Pre-salt field. The technology provides operational flexibility, as the decision regarding its application in one or more zones can be made in the brief window of time between well logging and the installation of the completion, without being constrained to field or well-specific wellhead design or configuration, thereby optimizing resources, improving operational efficiency and contributing to an increased Recovery Factor.

Acknowledgements

The authors would like to thank Petrobras and their partners for their permission to discuss the application presented in this paper.

Nomenclature

API	American Petroleum Institute
BOP	Blowout Preventer
CFD	Computational Flow Dynamics
CT	Coiled Tubing
eSSD	Electronic Sliding Side Door
FMECA	Failure Modes Effects Criticality Analysis
HPU	Hydraulic Pressure Unit
HSE	Health, Safety and Environment
ISO	International Organization for Standardization
NPT	Non-Productive Time
OBM	Oil-Based Mud
OPEX	Operating Expenses
PACI	Poco Aberto Completacao Inteligente (Portuguese for Open-Hole Intelligent Completion)
PCB	Printed Circuit Board
PDG	Permanent Downhole Gauge
POOH	Pull Out of Hole
RFID	Radio Frequency Identification
RIH	Run in Hole
Scf	Standard Cubic Feet
SIT	Systems Integration Testing
SPE	Society of Petroleum Engineers
TD	Total Depth
THP	Tubing Hanger Pressure

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